



Shri Vile Parle Kelavani Mandal's

DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING

(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



AI-Driven Clinical Decision Support Framework for Personalized Hereditary Cancer Risk Assessment

Guide: Prof. Adil Shaikh

Sanah Jhaveri D154 60009240013

Divya Kothari D088 60009240063

Dhir Shah D173 60009240035

Anish Nadar D009 60009230086

Table of Contents

- 1 Introduction
- 2 Literature Survey
- 3 Analysis of Literature Survey
- 4 Gaps in Existing Work
- 5 Problem Definition
- 6 Data Description
- 7 Proposed Design
- 8 Gap Resolution

Introduction

Hereditary cancers arise from inherited genetic variants that can substantially increase an individual's susceptibility to cancer. Advances in cancer genomics have made large-scale mutation and clinical datasets available for computational risk assessment. Conventional risk assessment often relies on individual cancer types, specific genes, or predefined clinical criteria, limiting the ability to capture complex interactions across multiple risk factors. Machine Learning provides an opportunity to integrate genomic variants with clinical characteristics and identify complex patterns associated with cancer risk.

An effective clinical decision-support framework should go beyond risk identification by supporting:

- overall lifetime cancer-risk estimation
- identification of relevant cancer types
- family-level risk assessment
- personalized screening and genetic-counselling support
- interpretable/explainable predictions

Literature Survey

Title	Objective	Approach / Methodology	Technologies / Algorithms	Dataset	Advantage
Prediction of Hereditary Cancers Using Neural Networks 2022	Predict hereditary cancer risk from family history and clinical data	Supervised deep learning using pedigree and clinical features	Artificial Neural Networks (ANN); family history features; clinical covariates	Hereditary cancer patient records; pedigree databases	Captures non-linear family-history patterns missed by rule-based models
Development and Validation of the PREMMplus Model for Multigene Hereditary Cancer Risk Assessment 2022	Extend PREMM model to predict risk for multiple Lynch syndrome genes simultaneously	Logistic regression model updated with multi-gene panel data	Logistic Regression; MLH1, MSH2, MSH6, PMS2, EPCAM gene features; clinical criteria	Multi-center hereditary cancer clinic cohorts	Broadens Lynch syndrome risk prediction to a full gene-panel scope
Insights on Variant Analysis in Silico Tools for Pathogenicity Prediction 2022	Evaluate and compare in silico pathogenicity prediction tools for germline variants	Systematic benchmarking of multiple in silico prediction algorithms	SIFT, PolyPhen-2, CADD, REVEL, SpliceAI; variant-level features; ensemble scoring	ClinVar; LOVD; hereditary cancer variant databases	Provides evidence-based guidance on tool selection for clinical variant interpretation
Predicting Cancer Risk from Germline Whole-Exome Sequencing Data Using a Novel Context-Based Variant Aggregation Approach 2023	Identify cancer predisposition variants from whole-exome sequencing data	ML classification applied to germline whole-exome sequencing features	Random Forest; gradient boosting; WES-derived variant features; gene burden scores	Germline WES cohorts from hereditary cancer patients and controls	Leverages full exome breadth to discover novel predisposition signals beyond known genes

Literature Survey

Title	Objective	Approach / Methodology	Technologies / Algorithms	Dataset	Advantages
Gene-Specific ML for BRCA1/2 Variants 2023	Classify pathogenicity of BRCA1/2 variants of uncertain significance	Supervised ML with gene-specific feature engineering	Random Forest, SVM; variant functional scores, conservation metrics, domain annotations	ClinVar, BRCA Exchange, functional assay data	Gene-specific model improves accuracy over generic pathogenicity predictors for BRCA1/2
Chatbot AI for Genetic Cancer Risk Assessment 2023	Automate patient-facing genetic cancer risk screening via conversational AI	NLP-driven chatbot integrated with risk assessment logic	Large Language Model (LLM); rule-based risk scoring; family history extraction	Patient self-reported family history data	Increases accessibility of genetic counseling; scalable patient intake without specialist bottleneck
Polygenic Scores in Cancer 2023	Evaluate utility of polygenic risk scores (PRS) for population-level cancer risk stratification	Meta-analytic review and PRS model benchmarking	Polygenic Risk Scores (PRS); GWAS summary statistics; logistic regression	UK Biobank, large-scale GWAS cohorts	Demonstrates PRS as complementary layer to monogenic risk; enables broader population screening
Deep Learning Framework for Complex Disease Risk Prediction 2023	Predict risk of complex diseases including cancer using multi-omics deep learning	End-to-end deep neural network on multi-omics inputs	Deep Neural Network (DNN); autoencoders; multi-omics feature fusion (genomic, transcriptomic)	TCGA, GEO multi-omics datasets	Integrates heterogeneous omics data; captures non-linear interactions missed by classical ML models
Disease-specific variant pathogenicity prediction (CardioBoost) Zhang et al., 2021	Show disease-specific classifiers outperform genome-wide tools	AdaBoost trained separately per disease (CM, IAS); ACMG-aligned probability thresholds	AdaBoost, 76 annotation features (incl. M-CAP/REVEL as inputs); high-confidence probability thresholds	SHaRe registry, ClinVar, HGMD, OMGL, gnomAD, DMS calmodulin data	First tool validated against patient survival/clinical outcome data

Literature Survey

Title	Objective	Approach / Methodology	Technologies / Algorithms	Dataset	Advantages
AI Techniques & Pedigree Charts in Oncogenetics 2024	Integrate AI with pedigree-based oncogenetic assessment	Hybrid pedigree + ML classification	Neural Networks + pedigree chart analysis; family history features	Clinical oncogenetics cohort	Combines traditional pedigree methods with modern AI for richer risk stratification
MAGPIE 2024	Predict pathogenicity of missense variants across genes	Gradient boosting with multi-gene feature integration	XGBoost; conservation, structural, and functional features	ClinVar, gnomAD, gene-specific variant databases	Generalizes across gene families; strong calibration for clinical use
Novel ML Pan-Cancer Risk Model 2024	Unified risk prediction across multiple cancer types	Multi-task ML framework trained on pan-cancer data	Random Forest + Logistic Regression ensemble; cross-cancer features	TCGA pan-cancer dataset	Single scalable model covering diverse cancer types simultaneously
ML Pancreatic Cancer Risk Prediction Using EHR 2024	Early detection of pancreatic cancer risk from EHR data	Supervised ML on structured clinical records	Gradient Boosting (XGBoost/LightGBM); EHR-derived clinical features	Large-scale EHR database	Leverages routinely collected clinical data; applicable in real-world settings
AI-Driven Mammography Future Breast Cancer Risk Prediction 2024	Predict future breast cancer risk from mammogram images	Deep learning image analysis on screening mammograms	CNN / Deep Learning; imaging biomarkers + density features	Screening mammography cohort	Non-invasive, image-based risk scoring integrated into existing screening workflow

Literature Survey

Title	Objective	Approach / Methodology	Technologies / Algorithms	Dataset	Advantages
Personalized Three-Year Survival Prediction 2024	Predict individualized 3-year cancer survival outcomes	Personalized ML survival modeling	Survival ML models (Cox-based + ensemble); clinical + genomic features	Multi-cancer clinical registry	Provides patient-specific survival estimates to guide treatment planning
ML-Based Cancer Risk Prediction Using Clinical & Demographic Factors 2024	Assess cancer risk using accessible clinical and demographic data	Supervised classification on tabular clinical data	Logistic Regression, Random Forest, SVM; demographic + clinical variables	Population-level clinical dataset	Low-cost, accessible approach suitable for resource-limited clinical settings

Analysis of Literature Survey

6

What existing research has established

1. Genomic information is valuable

- Pathogenic variants and mutation profiles can provide important information for hereditary cancer susceptibility.
- ML can identify complex relationships that may be difficult to capture using conventional statistical approaches.

2. Clinical information adds complementary predictive value

- Age, lifestyle, demographic and clinical variables can improve cancer-risk prediction.
- Pan-cancer studies demonstrate that combining multiple predictors can provide useful risk stratification.

3. AI can support hereditary cancer services

- AI-based tools have been investigated for genetic risk assessment and counselling.
- Existing studies demonstrate feasibility, but clinical validation and integration remain important challenges.

4. Research remains fragmented

- Genomic prediction, clinical risk assessment, family-risk assessment and counselling are frequently investigated as separate components.

Gaps In Existing Work

7

- Fragmented risk assessment: Existing approaches often focus on a specific cancer type, gene, or risk factor rather than integrating genomic, clinical, and family-history information into a unified assessment.
- Limited pan-cancer risk assessment: Many studies are designed for individual cancers or specific hereditary syndromes, limiting their ability to assess an individual's overall risk across multiple cancer types.
- Limited integration of family-level risk: Patient-level risk prediction and family/cascade risk assessment are often treated as separate processes, reducing the ability to directly translate individual risk into recommendations for relatives.
- Limited explainability and clinical translation: ML models can provide accurate predictions, but their outputs may not clearly explain the factors contributing to an individual's risk or directly translate predictions into personalized screening and genetic-counselling recommendations.

Problem Definition

8

Existing hereditary cancer risk-assessment approaches often evaluate genomic, clinical and familial risk factors independently or within cancer-specific models. This limits their ability to provide an integrated assessment of an individual's overall hereditary cancer risk and its implications across multiple cancer types. Furthermore, predicted risk is not always directly connected to family-level risk assessment, explainable decision-making and personalized clinical recommendations.

Therefore, there is a need for an integrated and explainable AI-driven clinical decision-support framework that combines genomic and clinical information to estimate overall lifetime hereditary cancer risk, identify relevant cancer-specific risks, incorporate family-level risk assessment, and support personalized screening and genetic-counselling decisions.

Data Description

9

1.Data Overview

Data Source:

- cBioPortal

Cancer Cohorts:

- Breast Invasive Carcinoma (BRCA) — BRCA1, BRCA2, TP53, PALB2, CHEK2, ATM, PTEN
- Ovarian Serous Cystadenocarcinoma (OV) — BRCA1, BRCA2, RAD51C, RAD51D, BRIP1, PALB2, TP53
- Pancreatic Adenocarcinoma (PAAD) — BRCA1, BRCA2, PALB2, ATM, CDKN2A, STK11, TP53
- Prostate Adenocarcinoma (PRAD) — BRCA1, BRCA2, ATM, CHEK2, PALB2, HOXB13, TP53
- Uterine Corpus Endometrial Carcinoma (UCEC) — MLH1, MSH2, MSH6, PMS2, EPCAM, PTEN, TP53
- Stomach Adenocarcinoma (STAD) — CDH1, CTNNA1, TP53, MLH1, MSH2, MSH6, PMS2
- Kidney Renal Clear Cell Carcinoma (KIRC) — VHL, PBRM1, BAP1, SETD2, TP53
- Brain Lower Grade Glioma (LGG) — IDH1, IDH2, TP53, ATRX, CIC, FUBP1
- Colorectal Cancer — APC, KRAS, TP53, BRAF, MLH1, MSH2, MSH6, PMS2, EPCAM

Data Description

11

Genomic / Mutation Data Includes:

- Patient ID
- Sample ID
- Gene Name
- Mutation Type -Describes the type of genetic alteration, such as SNV, insertion, deletion, or indel. It indicates how the DNA sequence has been altered.
- Variant Classification - Indicates the clinical significance of a genetic variant, such as pathogenic, likely pathogenic, benign, or uncertain significance. It helps determine whether the variant may be associated with disease risk.
- Protein Change - Describes the change produced in the protein sequence due to the genetic variant, usually represented as a change in amino acid(s). It helps assess the potential functional effect of the mutation.
- Other mutation-related attributes

Data Description

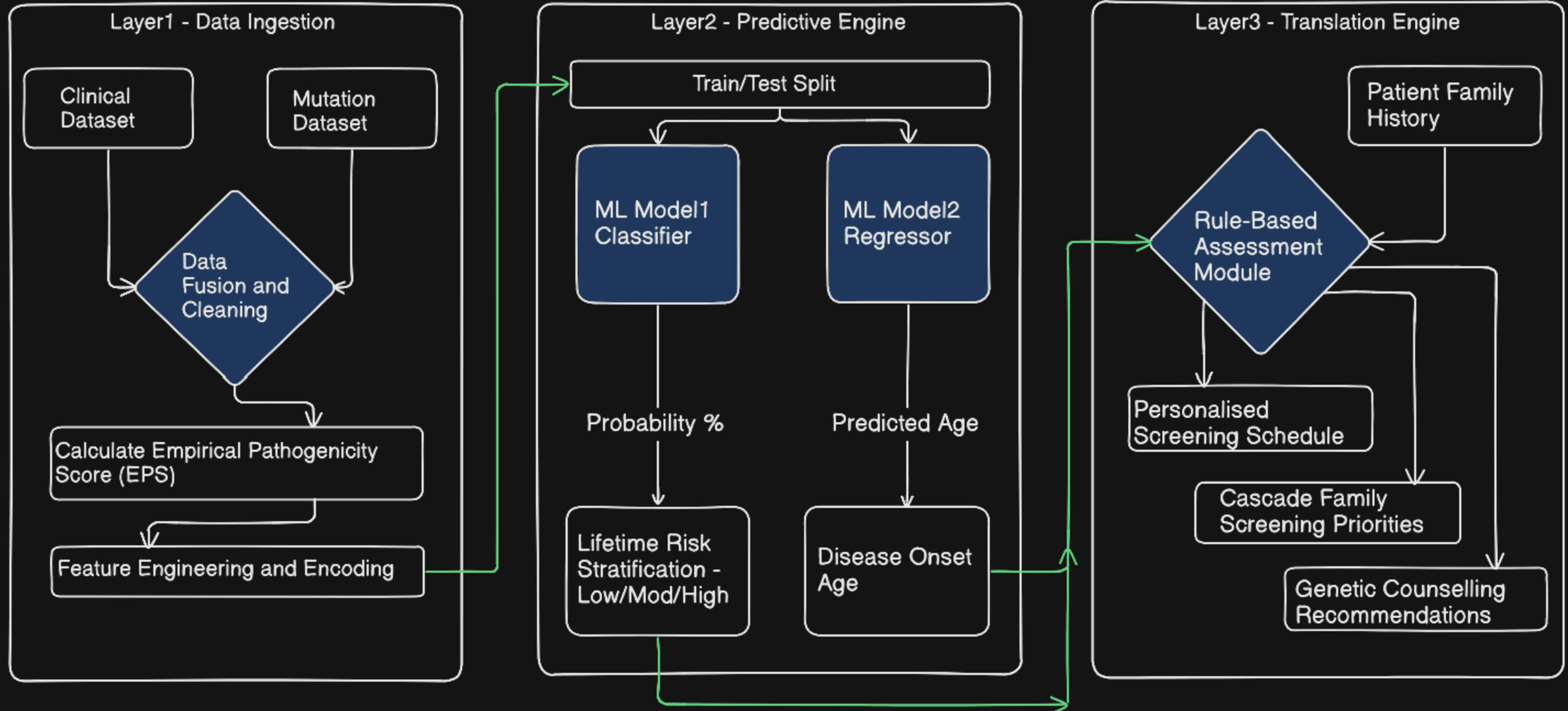
11

Clinical Data Includes:

- Age
- Gender
- Cancer Stage – Indicates the extent or progression of the cancer at the time of diagnosis.
- Survival Status – Indicates whether the patient was alive or deceased at the last recorded follow-up.
- Overall Survival Time – Represents the duration for which the patient survived after the defined starting point of the study.
- Tumor Site – Specifies the anatomical location where the tumor originated.
- Treatment Details – Provides information about the treatment received by the patient.
- Smoking History, where available

Proposed Design

12



Proposed Design

13

Layer 1 – Data Ingestion

- Clinical Dataset: Patient-related clinical information such as age, gender, cancer stage, survival status and lifestyle factors.
- Mutation Dataset: Genomic information including genes, mutation types, variant classification and molecular consequences.
- Data Fusion and Cleaning: Integrates the clinical and mutation datasets using the identified patient-level linking attribute, followed by data cleaning and preparation.
- Empirical Pathogenicity Score (EPS): Calculates a data-driven score to represent the relative pathogenic contribution of genomic variants.
- Feature Engineering and Encoding: Converts the integrated genomic and clinical information into suitable features for machine-learning models.

Proposed Design

14

Layer 2 – Predictive Engine

- Train/Test Split: Divides the prepared dataset into training and testing subsets for model development and evaluation.
- ML Model 1 – Classifier: Predicts the patient's overall lifetime cancer-risk probability and assigns the corresponding risk category.
- Lifetime Risk Stratification: Converts the predicted probability into risk categories such as Low, Moderate, High and Very High.
- ML Model 2 – Regressor: Estimates the likely age of disease onset based on the available genomic and clinical features.
- Predicted Age: Provides a temporal estimate that can support planning of appropriate surveillance.

Proposed Design

15

Layer 3 – Translation Engine

- Rule-Based Assessment Module: Combines the ML predictions with patient and family-history information using predefined clinical decision rules.
- Personalized Screening Schedule: Generates screening recommendations based on the patient's estimated risk and predicted disease-onset age.
- Cascade Family Screening Priorities: Identifies relatives who may require further genetic assessment or screening based on the patient's hereditary risk.
- Genetic Counselling Recommendations: Provides appropriate counselling recommendations based on individual and family-level risk.

Data Preparation

10

Data Parsing:

- Extraction of relevant information from the large cancer datasets.

Data Cleaning:

- Removal of duplicate records and handling of missing values.

Feature Selection:

- Identification of relevant genomic and clinical features.

Data Encoding:

- Conversion of categorical variables into numerical form.

Dataset Integration:

- Combining mutation datasets across the selected cancer cohorts.

Patient-Level Linking:

- Identification of a common patient identifier to connect the datasets.

Data Merging:

- Integration of genomic mutation data with corresponding clinical information.

Model Preparation:

- Preparation of the processed dataset for subsequent ML model development and evaluation.

Gap Resolution

16

Fragmented Risk Assessment Integrate genomic mutation data with clinical and family-history information.

- Provide a unified assessment instead of evaluating individual risk factors separately.

Limited Pan-Cancer Risk Assessment

- Integrate data across 9 cancer types.
- Estimate overall lifetime cancer-risk probability.
- Identify relevant cancer-specific risks when the overall risk is elevated.

Limited Family-Level Risk Integration

- Incorporate family history into the risk-assessment process.
- Generate cascade family screening priorities.
- Provide genetic counselling recommendations.
- Limited Explainability and Clinical Translation
- Incorporate Explainable AI (XAI) to improve interpretation of model predictions.
- Use a rule-based assessment module to translate predictions into understandable clinical recommendations.
- Generate personalized screening and follow-up recommendations.